Procedure

1. Open Android Studio IDE → select "Start a New Android Studio Project → specify the application name "UIWidgets" and company domain "com.uiwidgets" → click "next" → choose Minimum SDK "API 17:Android 4.2(Jelly Bean)" → click "Next" → choose "Blank Activity" → click "next" → specify the Activity Name "UIWidgets" → click "Finish".
2. Open UIWidgets.java under app/java/ uiwidgets.com. uiwidgets and type the following codes:

UIWidgets.java

```
package uiwidgets.com.uiwidgets;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.Toast;
import android.support.v7.app.AppCompatActivity;

public class UIWidgets extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_uiwidgets);

        // 1. Initialize Widgets
        final EditText textField = (EditText) findViewById(R.id.myTextField);
        final Spinner spinner = (Spinner) findViewById(R.id.mySpinner);
        Button button = (Button) findViewById(R.id.myButton);

        // 2. Populate the Spinner with data
        String[] options = {"Option A", "Option B", "Option C"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
                android.R.layout.simple_spinner_dropdown_item, options);
        spinner.setAdapter(adapter);

        // 3. Set Button Click Listener
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = textField.getText().toString();
                String selected = spinner.getSelectedItem().toString();

                Toast.makeText(UIWidgets.this,
                        "Text: " + input + "\nSelected: " + selected,
```

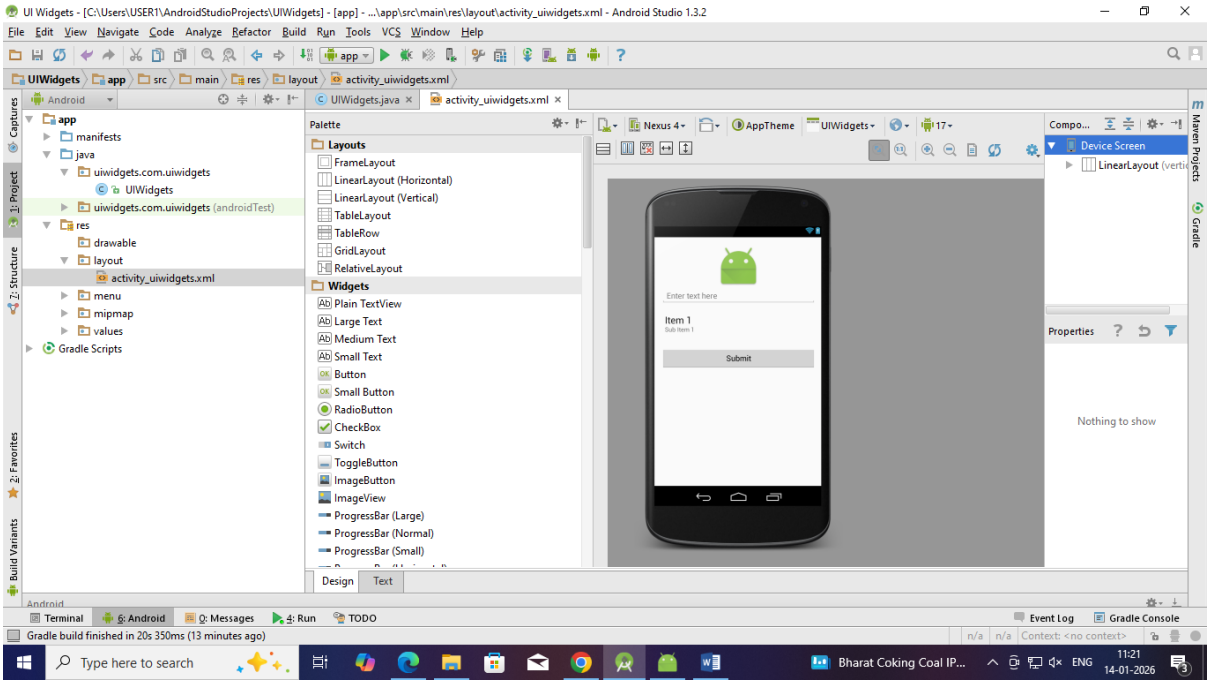
```
                Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

3. Open activity_uiwidgets.xml under app/res/layout and type the following codes:

activity_uiwidgets.xml

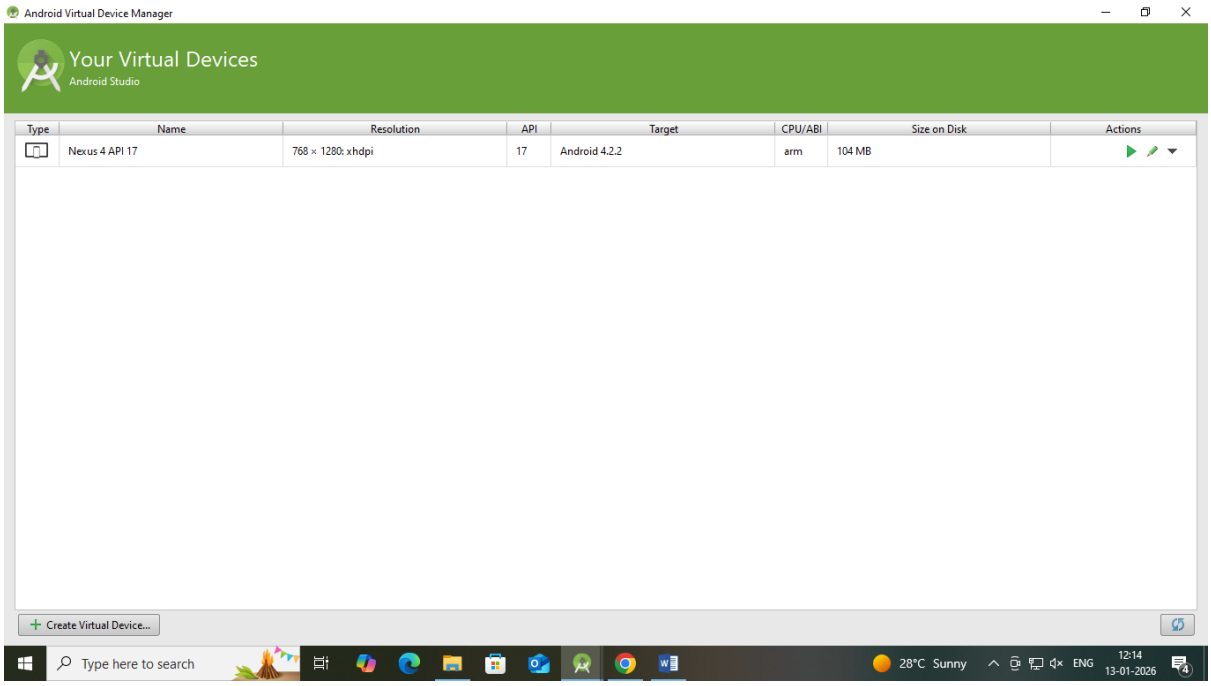
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <ImageView
        android:id="@+id/myImageView"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_gravity="center"
        android:src="@mipmap/ic_launcher" />

    <EditText
        android:id="@+id/myTextField"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text here" />

    <Spinner
        android:id="@+id/mySpinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/myButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Submit" />

</LinearLayout>
```

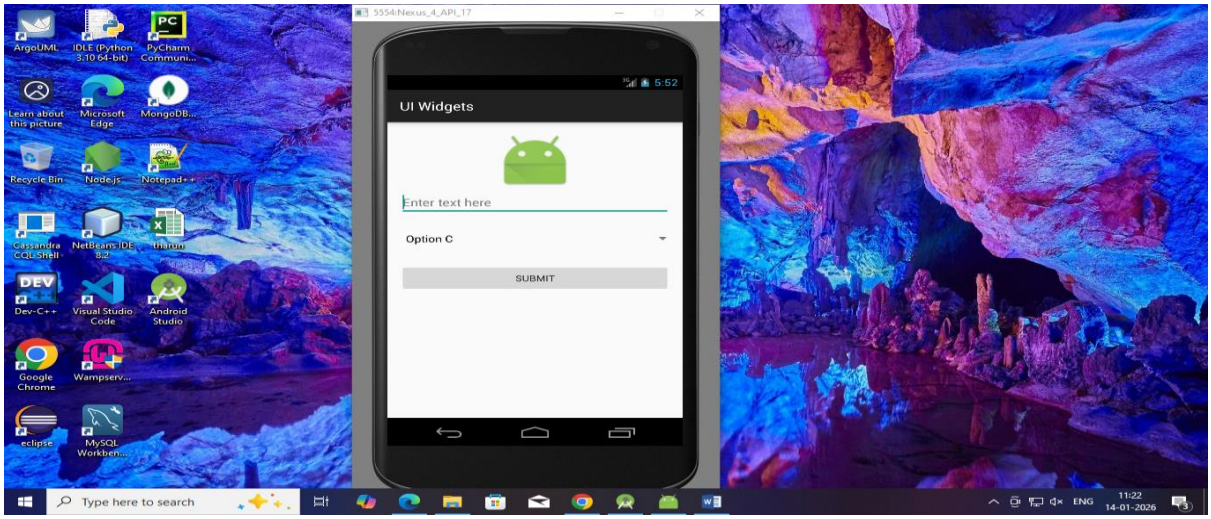
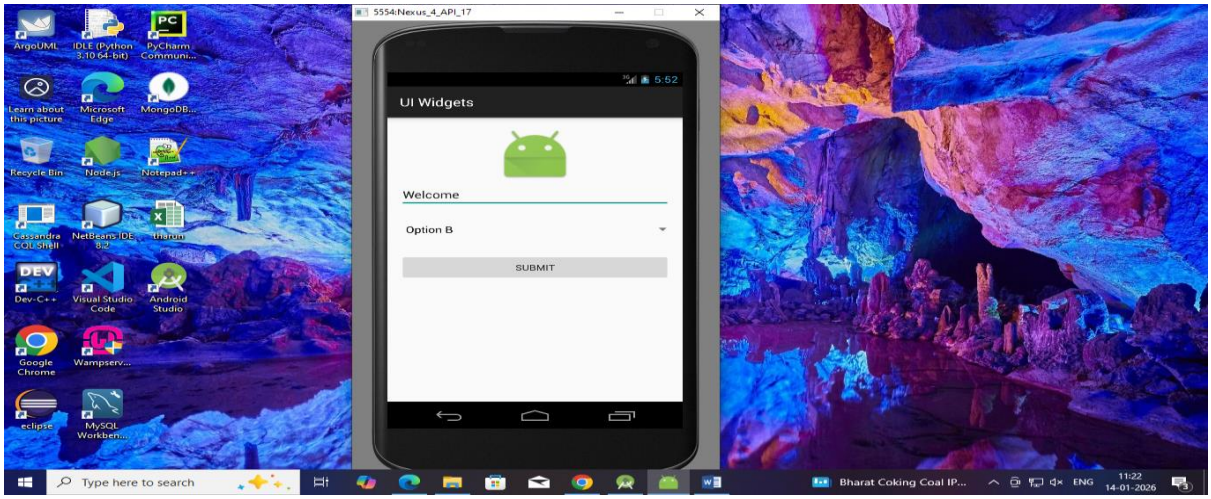
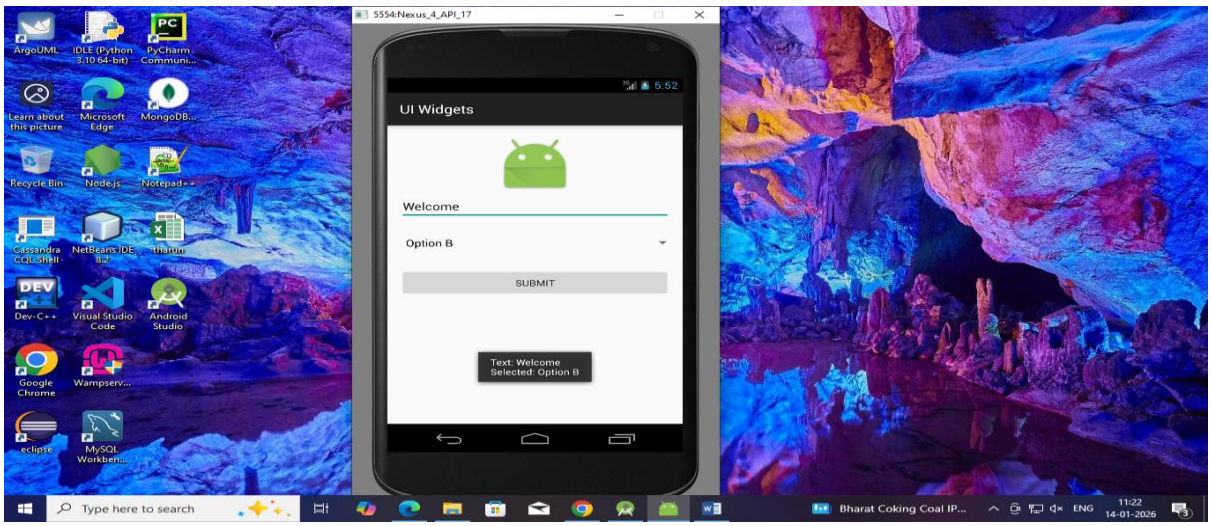
4. The design of the application will be as follows:



5. Go to Tools → android → AVD Manager → click "+ create a virtual device" → select "phone" from category → select "Nexus 4" from the list → click "next" → select Release name: Jelly Bean, API Level: 17, ABI: armeabi-v7a, Target: Android 4.4.2 from the list → click "next" → Choose orientation "portrait" → click "finish".

The following window will appear after configuring AVD:



6. Click “Run app” button in the Android Studio → choose android virtual device → click “ok”.

Output